

Social engineering is a form of cyber attack that focuses on manipulating people rather than exploiting technical vulnerabilities. Instead of breaking into systems directly, attackers use deception, persuasion, and psychological tactics to gain access to information, systems, or physical locations. This makes social engineering one of the most effective and widely used methods in cyber security incidents.

At its core, social engineering relies on human behaviour. Attackers take advantage of trust, curiosity, fear, or a sense of urgency to influence decisions. By impersonating trusted individuals or organisations, they can convince users to reveal sensitive information, perform actions, or bypass normal security procedures.

Social engineering can take several forms, often overlapping with other types of attacks:

- **Phishing** – deceptive messages designed to trick users into revealing information
- **Pretexting** – creating a fabricated scenario to obtain data or access
- **Baiting** – offering something enticing, such as free downloads, to lure users into a trap
- **Tailgating** – gaining physical access by following authorised individuals into restricted areas
- **Impersonation** – pretending to be a trusted person, such as an IT employee or manager

These techniques are effective because they exploit natural human tendencies rather than technical weaknesses.

Social engineering attacks typically follow a structured approach. An attacker will first gather information about their target, such as job roles, company structure, or personal details. They then use this information to craft a believable scenario. Once contact is made, the attacker attempts to build trust or create pressure, encouraging the target to act without fully verifying the situation.

For individuals, social engineering can lead to the loss of personal data, financial fraud, or unauthorised access to accounts. Because these attacks often appear legitimate, users may not realise they have been targeted until after the damage has occurred.

For businesses, the consequences can be more severe. Social engineering is frequently used to bypass security controls, leading to data breaches, financial loss, and operational disruption. Attackers may target employees with access to sensitive systems or information, making organisations particularly vulnerable if staff are not properly trained.

Recognising social engineering attempts is essential for reducing risk. Common warning signs include:

- Requests for sensitive information that are unusual or unexpected
- Messages or calls that create urgency or pressure to act quickly
- Individuals refusing to follow normal verification procedures
- Inconsistencies in communication, such as incorrect details or unfamiliar contact information
- Offers that seem too good to be true or requests that seem out of place

Being able to identify these indicators can help prevent attackers from gaining access.

Protecting against social engineering requires a combination of awareness and structured processes:

- Verifying identities before sharing information or granting access
- Following established procedures, even under pressure
- Avoiding the disclosure of sensitive information without proper authorisation
- Providing regular training and awareness programmes within organisations
- Encouraging a culture where users feel comfortable questioning unusual requests
- Implementing policies for handling sensitive data and access requests

For organisations, additional measures such as access controls, monitoring, and clear reporting procedures can further reduce the risk of successful attacks.

Social engineering remains a significant threat because it targets the human element of security. Even well-protected systems can be compromised if individuals are persuaded to bypass controls. Understanding how these attacks work, how to recognise them, and how to respond appropriately is essential for both individuals and organisations.

This page provides an overview of social engineering, including its methods, risks, and preventative measures. Developing awareness of these tactics is a key step in strengthening overall cyber security.